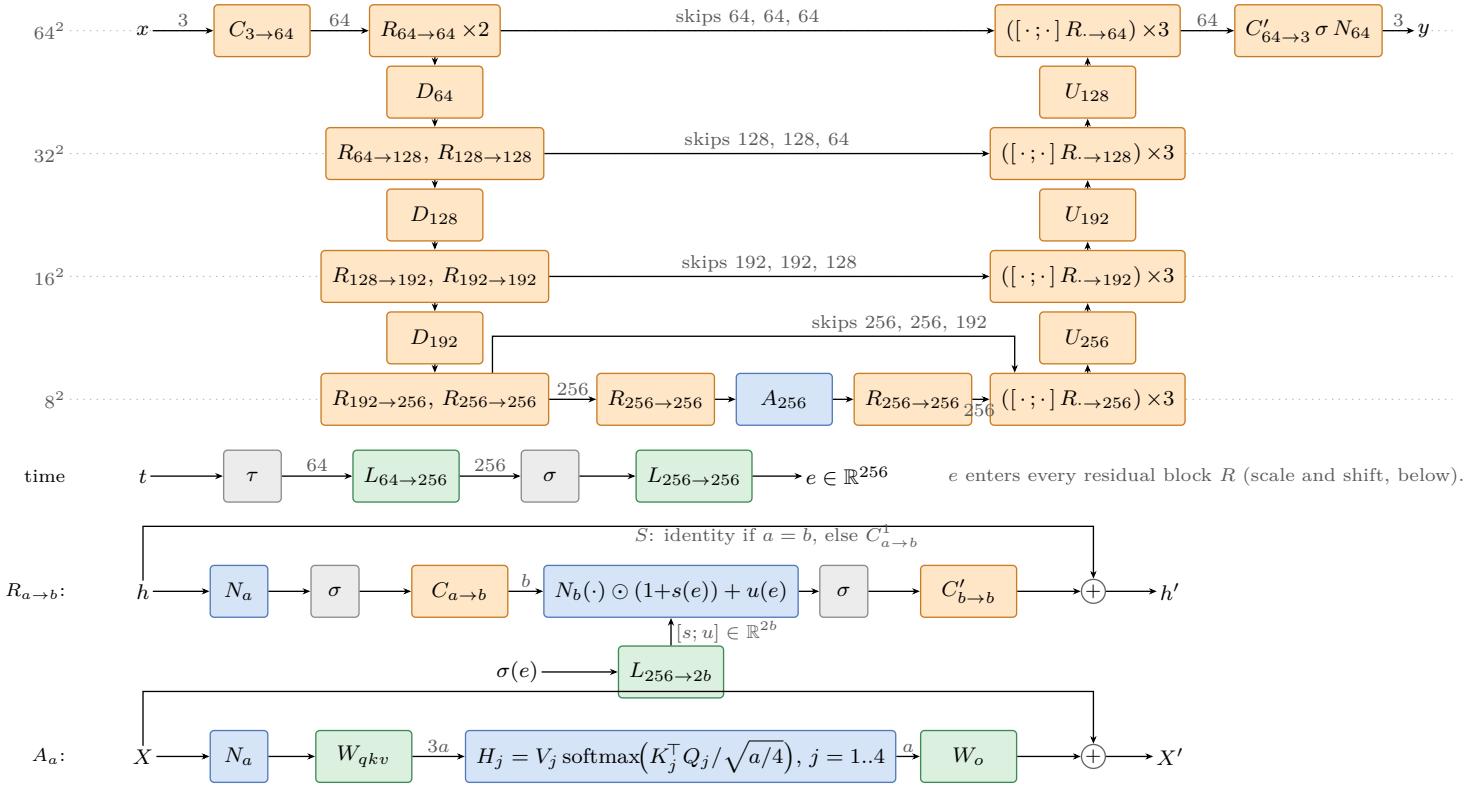
MODEL CARD the U-Net of the reference code for CelebA at 64×64: $\varphi_\theta : \mathbb{R}^{3\times64\times64} \times \mathbb{R} \to \mathbb{R}^{3\times64\times64}$, 20 972 867 parameters.



□ per-pixel affine map: a sparse linear layer with fan-in K (dense when $K = n_{\text{in}}$) □ 3×3 convolution: parameter-free unfold, then a per-pixel sparse linear map $\mathbb{R}^{9a} \to \mathbb{R}^b$ with shared, structured support □ substitutable: softmax attention (by attention-free or windowed causal attention), group norm (by dynamic tanh or RMS norm) □ parameter-free

FIGURE 1. Lanes are the map sizes, edge labels the channel counts (Table 1 defines the symbols). Each decoder block concatenates the tensor arriving from below with one encoder output of its lane (the three skips of a lane pair in reverse order with the encoder outputs of that lane). In A_a the $m = hw = 64$ pixels are the tokens, $[Q;K;V] = W_{qkv}N_a(X) + b$, the softmax runs over the key index. $\tau(t) = (\cos\omega_i t)_{i<32} \oplus (\sin\omega_i t)_{i<32}$, $\omega_i = 10^{-4i/32}$: the positional table of a transformer at the real argument t. A composite box takes the colour of its weight-carrying parts.

stage	operation	map	channels	parameters
input	$x \in \mathbb{R}^{3\times64\times64}$, pixels in $[0,1]$	64^2	3	
time	$e = L_{256\to256}\,\sigma\,L_{64\to256}\,\tau(t)$		$64 \to 256 \to 256$	82 432 □
stem	$h = C_{3\to64}\,x$	64^2	$3 \to 64$	1 792 □
encoder	$R_{64\to64},\ R_{64\to64}$	64^2	$64 \to 64 \to 64$	214 016 □
	$h \leftarrow D_{64}h$	$64^2 \to 32^2$	64	36 928 □
	$R_{64\to128},\ R_{128\to128}$	32^2	$64 \to 128 \to 128$	657 408 □
	$h \leftarrow D_{128}h$	$32^2 \to 16^2$	128	147 584 □
	$R_{128\to192},\ R_{192\to192}$	16^2	$128 \to 192 \to 192$	1 440 832 □
	$h \leftarrow D_{192}h$	$16^2 \to 8^2$	192	331 968 □
	$R_{192\to256},\ R_{256\to256}$	8^2	$192 \to 256 \to 256$	2 527 360 □
middle	$R_{256\to256}$	8^2	256	1 312 768 □
	A_{256}, 4 heads of 64 channels	8^2	$256 \to 768 \to 256$	263 680 □
	$R_{256\to256}$	8^2	256	1 312 768 □
decoder	$([\cdot;\cdot]\,R_{\cdot\to256})\times3$	8^2	$512, 512, 448 \to 256$	5 939 328 □
	$h \leftarrow U_{256}h$	$8^2 \to 16^2$	256	590 080 □
	$([\cdot;\cdot]\,R_{\cdot\to192})\times3$	16^2	$448, 384, 320 \to 192$	3 508 416 □
	$h \leftarrow U_{192}h$	$16^2 \to 32^2$	192	331 968 □
	$([\cdot;\cdot]\,R_{\cdot\to128})\times3$	32^2	$320, 256, 192 \to 128$	1 626 240 □
	$h \leftarrow U_{128}h$	$32^2 \to 64^2$	128	147 584 □
	$([\cdot;\cdot]\,R_{\cdot\to64})\times3$	64^2	$192, 128, 128 \to 64$	497 856 □
head	$y = C'_{64\to3}\,\sigma\,N_{64}\,h$	64^2	$64 \to 3$	1 859 □
total				20 972 867

architecture	
widths	$64\cdot(1,2,3,4)$ at $64^2, 32^2, 16^2, 8^2$
blocks per level	2 (encoder), 3 (decoder)
attention	middle block only, 4 heads
normalisation	group norm, 32 groups
nonlinearity	SiLU
dropout	0
conditioning	scale and shift from $e \in \mathbb{R}^{256}$
zero-initialised	C' in every R, W_o, head conv
diffusion and training (CelebA config)	
steps	$N = 50$
step sizes	γ_k: linear ramp $10^{-5} \to 0.1$, mirrored; $T = \sum_k \gamma_k = 2.50$
prior	$\mathcal{N}(m\mathbf{1}, 0.5\,I)$, m = mean pixel of 100 images
loss	mean matching, $\lvert\varphi(x_{k+1}, T{-}t_k) - x_{k+1} - \text{target}_k\rvert^2$
optimiser	Adam, 10^{-4}, gradient norm clipped at 1
EMA	0.999 (used for simulation and sampling)
batch	128
cache	300 trajectories × 50 steps, refreshed every 100 steps
IPF	20 iterations × 50 000 steps per half-step
data	centre crop 140, resize 64, random flip

TABLE 1. Stages in order of execution. $R_{a\to b}(h,e) = S(h) + C'_{b\to b}\,\sigma\big(N_b(C_{a\to b}\,\sigma\,N_a h)\odot(1+s(e))+u(e)\big)$ with $[s(e);u(e)] = L_{256\to2b}\,\sigma(e) \in \mathbb{R}^{2b}$ broadcast over pixels and S the identity when $a = b$, else $C^1_{a\to b}$ ($2a + 9ab + 9b^2 + 518b + [a \neq b](ab + b)$ parameters). $A_a(X) = X + W_o[H_1;\ldots;H_4] + b_o$ ($4a^2 + 6a$). $C_{a\to b}$: $9ab + b$; D_a, U_a: $9a^2 + a$; $L_{a\to b}$, $C^1_{a\to b}$: $ab + b$; N_a: $2a$.

FAN-IN. A sparse linear layer $S : \mathbb{R}^{n_{\text{in}}} \to \mathbb{R}^{n_{\text{out}}}$ with fan-in K computes $S(x) = (M \odot W)x + b$, $M \in \{0,1\}^{n_{\text{out}}\times n_{\text{in}}}$ with K ones per row drawn uniformly without replacement ($1 \le K \le n_{\text{in}}$; $K = n_{\text{in}}$ is the dense map; the masked dense weight and the $n_{\text{out}} \times K$ weight with an index table are two implementations of the same map). K fixes in-degrees, not the maximum degree K_G of the network graph: with $k_i^- = \sum_j M_{ij} = K$ for every output unit i and $d_j^+ = \sum_i M_{ij}$ for every input unit j,

$$\mathbb{E}\,d_j^+ = \frac{n_{\text{out}}K}{n_{\text{in}}}, \qquad k_v = K_{l(v)} + d_v^+ \text{ for a hidden unit } v \text{ of layer } l(v), \qquad K_G := \max_v k_v \;\ge\; \max_l K_l,$$

with equality only for supports without reuse, which random supports do not enforce. The fan-in K (per layer, K_l) is the quantity of interest; a green box is such a map for any $K \le n_{\text{in}}$.